INCH-POUND

MIL-T-49303(CR)
18 August 1988

MILITARY SPECIFICATION

TRANSPONDER SET AN/PPN-19(V)

This specification is approved for use within the Communications-Electronics Command, Fort Monmouth, New Jersey, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for construction and testing of Transponder Set AN/PPN-19(V) (Multifunction Radar Transponder Beacon). AN/PPN-19(V) is a self-contained lightweight, man portable unit designed for field service, and shall consist of:

Transponder Set AN/PPN-19(V)1 Configuration

Receiver - Transmitter, Radar, RT-1493/PPN-19(V) C5083053
Antenna, AS-3779/PPN-19(V) C5083043
Switch, Waveguide, SA-2495/PPN-19(V) C5083033
Control, Transponder Set, C-11601/PPN-19(V) C5083321

Transponder Set AN/PPN-19(V)2 Configuration

Receiver, Transmitter, Radar, RT-1493/PPN-19(V) C5083053
Antenna, AS-3779/PPN-19(V) C5083043
Switch, Waveguide, SA-2495/PPN-19(V) C5083033
Control, Transponder Set, C-11601/PPN-19(V) C5083321
Battery Box, CY-8412/PPN-19(V) C5083329

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC - N/A FSC 5895
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SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes for Ground Electronic Equipment.
MIL-P-15024/6	- Plate, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-T-28800	- Test Equipment for use with Electrical and Electronic Equipment, General Specification for.
MIL-E-55585	- Electronics Equipment and Parts, Packaging of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-781	- Reliability Tests Exponential Distribution.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-2073-2	- Packaging Requirement Codes

2.1.2 Other government documents and drawings. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

DLC5083030	- Transponder Set Assembly, AN/PPN-19.
C5083031	- ATP, Switch Cap
C5083033	- Switch Cap Assembly
C5083043	- Omnidirectional Antenna Assembly

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C5083053 - Receiver/Transmitter Assembly
C5083058 - Signal Processor Assembly
C5083059 - ATP, Signal Processor Unit
C5083106 - ATP, MPU
C5083112 - ATP, Encoder
C5083119 - ATP, Panel Interface
C5083121 - Control Panel Assembly
C5083126 - Power Supply Assembly
C5083151 - ATP, RF Cable
C5083201 - ATP, I-Band RF Head
C5083214 - ATP, J-Band RF Head
C5083227 - ATP, Bit Source
C5083238 - ATP, J-Band MAG/MOD
C5083259 - ATP, I-Band MAG/MOD
C5083263 - ATP, Control Panel
C5083266 - ATP, I-Band IF
C5083276 - ATP, I-Band IF Amplifier
C5083278 - K-Band Video Assembly
C5083279 - ATP, K-Band Video
C5083293 - ATP, J-Band Video
C5083306 - ATP, J-Band FET Amplifier
C5083321 - Remote Control Unit Assembly
C5083322 - ATP, Remote Control Unit
C5083329 - Battery Box Assembly
C5083333 - ATP, Battery Protection
C5083347 - ATP, Power Supply
C5083449 - ATP, K-Band Detector
C5087088 - ATP, I-Band Preselector
C5087090 - ATP, J-Band Preselector
C5087094 - ATP, RF Transition
C5087096 - ATP, J-Band Filter/Det
C5087098 - ATP, I-Band Horn
C5087100 - ATP, J-Band Horn
C5087102 - ATP, K-Band Horn
C5087104 - ATP, DRO
C5087106 - Preliminary Alignment and Test Procedure
C5087108 - Final Acceptance Test Plan (ATP).
C5087110 - ATP, Battery Box Adaptor

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (escept for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specified exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish first article units for first article inspection and approval (see 4.3 and 6.2).

3.2 Construction. The Transponder Set AN/PPN-19(V) shall be constructed in accordance with the requirements of this specification, DLC5083030, and the following list of major components:

Component	Drawing
Switch Cap Assembly	C5083033
Omnidirectional Antenna Assembly	C5083043
Receiver/Transmitter Assembly	C5083053
Signal Processor Assembly	C5083058
Control Panel Assembly	C5083121
Power Supply Assembly	C5083126
Remote Control Unit Assembly	C5083321
Battery Box Assembly	C5083329

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply (see 4.4).

3.3.1 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Finish. All surfaces of the AN/PPN-19(V) requiring a protective coating shall be finished in accordance with MIL-F-14072 and the equipment drawings specified herein. (See 4.4).

3.5 Nameplate and marking.

3.5.1 General. Marking of AN/PPN-19(V) shall be in accordance with MIL-M-13231 (see 4.4.1).

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.5.3 Nameplates. Each nomenclature item shall have a nameplate conforming to MIL-P-15024/6, Type H or B (see 4.4).

3.5.4 Serial numbers. Each AN/PPN-19(V) shall be serial numbered.

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3.6 Safety engineering.

3.6.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-T-28800 shall be made applicable for tasks and efforts such as the selection of parts, the complete manufactuing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (See 4.13).

3.6.2 Edge rounding exposed. Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners.

3.6.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.7 Interchangeability.

3.7.1 Mechanical interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically interchangeable without modification and shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.7.2 Electrical interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be functionally interchangeable without modification and shall conform to Requirement 7 of MIL-STD-454. Electrical adjustment may be made at all maintenance levels as referenced in the manuals (see 4.8).

3.8 Dimensions and weights. The weight of receiver-transmitter shall not exceed 17.75 lbs (8.0 kg) excluding battery box. The dimensions of the receiver-transmitter shall be 12.75 x 4.75 x 8.68 inches (32.4cm x 12cm x 22cm) excluding protrusions and battery box. The weight of battery box, excluding batteries, shall not exceed 4.5 lbs (2.0kg). The dimensions of the battery box shall be 12.75 x 5.1 x 5.0 inches (32.4 cm x 13 cm x 12.7 cm) excluding protrusions (see 4.7).

3.9 Environmental stress screening (ESS). The AN/PPN-19(V) shall meet the performance requirements specified herein after subjection to vibration and electrical preconditioning test of 4.6.

3.10 Performance requirements. The AN/PPN-19(V) shall meet the performance requirements as specified in 3.2 after subjection to the test of 4.9.

3.11 Service conditions. The AN/PPN-19(V) shall meet the following service conditions (See 4.10).

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3.11.1 Altitude. The AN/PPN-19(V), including battery box shall meet specified performance, without degradation, at altitudes up to and including 10,000 feet (3048m) above sea level, and shall withstand air transportation at altitudes up to 40,000 feet (12,192m) above sea level (see 4.10.1).

3.11.2 High temperature. The AN/PPN-19(V), including battery box shall be operable without deterioration damage or degradation in specified performance at ambient air temperature as high as +125°F (51.67°C), plus the effect of solar radiation at a rate of 360 BTU/ft^2/hr. The AN/PPN-19(V) shall be permitted a one minute cool-down period before full operation is required. At the end of the one minute period, the AN/PPN-19(V) shall provide full specified performance. The AN/PPN-19(V), including battery box, shall be capable of withstanding without degradation of specified performance long periods of storage at temperature as high as +160°F (71.1°C) (see 4.10.2).

3.11.3 Low temperature. The AN/PPN-19(V), including battery box, shall be operable without deterioration damage or degradation in specified performance at air temperature as low as -50°F (-45.6°C). The AN/PPN-19(V), including battery box, shall be capable of withstanding, without degradation of specified performance, long periods of storage at temperatures as low as -70°F (-56.7°C) (see 4.10.3).

3.11.3.1 Warm-up time. The receivers, transmitters, power supply, and associated circuitry shall become fully stabilized within the system operational parameters specified herein in a time period not to exceed one minute under all operational service conditions specified herein, except below -10°C. An increased time shall be allowed for frequency stability in a linear manner from one minute at -10°C up to five minutes at -46°C. Stabilization time is measured from the time of application of the specified system input voltages.

3.11.4 Humidity. The AN/PPN-19(V), including battery box, shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme humidity levels as encountered in tropical areas (see 4.10.4).

3.11.5 Salt fog. After the salt fog test of 4.10.5, the AN/PPN-19(V), including battery box shall operate without degradation in specified performance and show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface exfoliations. There shall also be no clogging or binding of moving mechanical parts.

3.11.6 Fungus. After the fungus test of 4.10.6, the AN/PPN-19(V), including battery box, shall be visually examined. The AN/PPN-19(V) shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area, and no more than six unrelated minuted colonies with mycelial development in only areas

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other than critical circuit portions; such as terminal spacing, printed curcuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the AN/PPN-19(V) shall operate without degradation in specified performance after exposure to the test conditions of 4.10.6. Materials used in the manufacture, fabrication, construction and treatment of the AN/PPN-19(V) shall comply with Requirement 4 of MIL-STD-454.

3.11.7 Rain. The AN/PPN-19(V), including battery box, shall be operable without degradation in specified performance, and shall sustain no physical damage, during period of heavy precipitation with intermittent wind. (See 4.10.7)

3.11.8 Immersion. The AN/PPN-19(V), including battery box, shall be immersion proof to a covering depth of 3 feet of water for a period of not less than 2 hours. (See 4.10.8).

3.11.9 Wind. The AN/PPN-19(V), including battery box, shall be operable and sustain no physical damage during the winds of 45 knots for a 5 minute period with gust to 65 knots. (See 4.10.9).

3.11.10 Dust. The AN/PPN-19(V), including battery box, shall withstand, in both operating and non-operating condition, exposure to fine dust particles with wind speed of 17 knots. (See 4.10.10.)

3.11.11 Vibration (loose cargo). The AN/PPN-19(V) shall operate within specified performance after being subjected to the vibration test specified in 4.10.11.

3.11.12 Vibration (secured cargo). The AN/PPN-19(V) shall operate within specified performance after being subjected to the vibration test specified in 4.10.12.

3.11.13 Shock. The AN/PPN-19(V) shall operate within specified performance after being subjected to the method and condition of the test specified in 4.10.13.

3.11.14 Bench handling. The AN/PPN-19(V) shall withstand the shock encountered during servicing and operate within specified performance. (See 4.10.14).

3.12 Electromagnetic interference (EMI). The equipment shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4: (See 4.11).

CE01	CS01	RE02	(a)RS03
CE04	CS02	RE02.1	
	CS06		

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(a) The equipment shall be immersed in electric fields from 10 kHz to 10 GHz having the following intensities and modulation characteristics:

Frequency	Intensity	Modulation
10 kHz - 1.9 MHz	1 V/M	AM, 50%, 1 kHz tone
2.0 MHz - 22.99 MHz	5 V/M	AM, 50%, 1 kHz tone
30 MHz - 75.9 MHz	10 V/M	FM, ±8 kHz deviation, 1 kHz tone
76 MHz - 399.9 MHz	10 V/M	AM, 50%, 1 kHz tone
400 MHz - 1999 MHz	10 V/M	Pulse, 0.1 usec 400 pps
2.0 GHz - 10 GHz	5 V/M	Pulse, 0.1 usec 400 pps

3.13 Reliability. (See 4.12). The AN/PPN-19(V) shall have a specified (O_o) mean-time-between-failure (MTBF) as defined in MIL-STD-781 of 600 hours. The lower test MTBF (O_1) as defined in MIL-STD-781 shall be 200 hours. The reliability requirements shall be demonstrated when operated in accordance with the following conditions:

a. Ambient temperature: -46°C to +68°C.

b. Ground mobile environment.

c. Transmit 5 minutes out of every 10 hours of uninterrogated operation in standby power mode.

3.14 Workmanship. The AN/PPN-19(V) shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9, and the applicable portions of the following paragraphs of MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.
Cleaning of units.
General requirements for securing parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is reponsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection of equipment before packaging (see 4.5).
d. Packaging inspection (see 4.14).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish nine first article units of AN/PPN-19(V) for group A, B, C and D testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspection specified in table I and shall be performed in the order specified.

4.3.3 First article test. The first article test plan and test reports shall be as required in the contract.

4.4 Inspection covered by subsidiary document. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	When required
Parts, materials, and process; general	3.3
Finish.	3.4
Nameplate and marking.	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4 and 4.6. This does not relieve the contractor of his responsibility

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TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of Test Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6-9
1. Inspections covered subsidiary documents 1/	3.3 3.4 3.5	4.4	Inspections to be performed on all units					
2. Environmental stress screening 1/	3.9	4.6	Tests to be performed on all units					
3. Group A. inspection 1/	See table II		Tests to be performed on all units					
4. Group B inspection 1/	See table III		Tests to be performed on all units					
5. Group C inspection								
Altitude	3.11.1	4.10.1		1				
Low temperature	3.11.3	4.10.3	1					
High temperature	3.11.2	4.10.2	2					
Humidity	3.11.4	4.10.4	3					
Salt fog 2/	3.11.5	4.10.5				1		
Fungus 2/	3.11.6	4.10.6					1	
Rain	3.11.7	4.10.7		4				
Immersion	3.11.8	4.10.8			1,6			
Wind	3.11.9	4.10.9		2				
Dust	3.11.10	4.10.10		3				
Vibration(loose cargo)	3.11.11	4.10.11			2			
Vibration(secured cargo)	3.11.12	4.10.12			3			
Shock	3.11.13	4.10.13			4			
Bench handling	3.11.14	4.10.14			5			
6. EMI	3.12	4.11		5				
7. Group D inspection								
Reliability	3.13	4.12						1

1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ The salt and fungus tests shall be performed only on the units selected from the first group of 50 units during the first article inspection.

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for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and acceptable quality levels indicated in table II.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL major	AQL minor
Visual and mechanical	3.14	4.13,4,4	1%	4%
System safety	3.6	4.13	1%	4%
Performance	3.10	4.9	1%	1/

1/ All electrical defects are major.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to meet the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sample plans, for the AQL's listed in table III, shall be inspection level S-1.

4.5.2.2 Order of inspection with group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Interchangeability	3.7	4.8	6.5%
Dimensions and weights	3.8	4.7	6.5%

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4.5.3 Group C inspection. Group C inspection shall be performed on inspection lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Low temperature	3.11.3	4.10.3
High temperature	3.11.2	4.10.2
Humidity	3.11.4	4.10.4
Subgroup 2		
EM I	3.12	4.11
Altitude	3.11.1	4.10.1
Wind	3.11.9	4.10.9
Dust	3.11.10	4.10.10
Rain	3.11.7	4.10.7
Subgroup 3		
Immersion	3.11.8	4.10.8
Vibration loose cargo	3.11.11	4.10.11
Vibration secured cargo	3.11.12	4.10.12
Shock	3.11.13	4.10.13
Bench handling	3.11.14	4.10.14

4.5.3.1 Sampling for group C inspection. The AN/PPN-19(V) shall be selected at random, one sample for each subgroup 1, 2 and 3 in Table IV, from every 150 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot. The salt fog and fungus tests shall be performed only on the first article units selected from the first group of 50 units.

4.5.3.2 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and have passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

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4.5.4 Group D inspection. This inspection shall consist of the test specified in table V and shall be performed on units from lots which have been subjected to and have met group A and group B inspections.

4.5.4.1 Sampling for group D inspection. Two samples of AN/PPN-19(V) shall be selected at random for group D inspection from every 150 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot.

TABLE V. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.13	4.12

4.5.4.2 Group D failures. In case of group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.4.3 Reinspection of group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

4.6 Environmental stress screening. Prior to Group A inspection, each AN/PPN-19(V) shall be operated for a maximum of 48 hours under the conditions specified in 4.12 with the following exceptions: test cycle shall be 8 hours long and the system shall be operated at nominal voltage 100 percent of the time. The burn-in shall be concluded by a failure free period, the duration of which shall be 16 hours. The minimum performance parameters specified in 4.12.3.1 shall be measured and recorded daily.

The burn-in test procedures and acceptance test conditions are defined in the following list of drawings:

Preliminary Alignment and Test Procedure	C5087166
Final Acceptance Test Procedure	C5087108
Module Burn-in	C5087119
System Burn-in	C5087211

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4.7 Dimensions and weights. The AN/PPN-19(V), including battery box, shall be measured and weighed on a calibrated scale to determine compliance with dimensions and weight requirements of 3.8.

4.8 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the units. Mechanical dimensions and electrical performance required by the AN/PPN-19(V) drawings shall be measured to determine compliance with Requirement 7 of MIL-STD-454 (see 3.7).

4.9 Performance test. The AN/PPN-19(V) shall be subjected to performance test and electrical inspections in accordance with the following list of ATP drawings (see 3.10).

C5083031	Switch Cap
C5083059	Signal Processor Unit
C5083106	MPU
C5083112	Encoder
C5083119	Panel Interface
C5083151	RF Cable
C5083201	I-Band RF Head
C5083214	J-Band RF Head
C5083227	Bit Source
C5083238	J-Band MAG/MOD
C5083259	I-Band MAG/MOD
C5083263	Control Panel
C5083266	I-Band IF
C5083276	I-Band IF Amplifier
C5083279	K-Band Video
C5083293	J-Band Video
C5083306	J-Band FET Amplifier/Det.
C5083322	Remote Control Unit
C5083333	Battery Protection
C5083347	Power Supply
C5083449	K-Band Detector
C5087088	I-Band Preselector
C5087090	J-Band Preselector
C5087094	RF Transition
C5087096	J-Band Filter/Det
C5087098	I-Band Horn
C5087100	J-Band Horn
C5087102	K-Band Horn
C5087104	I-Band DRO
C5087106	Preliminary Alignment and Test Procedure
C5087108	Final Acceptance Test (ATP)
C5087110	Battery Box Adapter

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4.10 Service conditions test. Service condition tests shall be performed as detailed below. Each equipment subjected to these tests shall have passed group A tests and shall be resubmitted to and shall pass these group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass, as a minimum, the "bench test parameters" listed in table VI.

TABLE VI. Bench test to be performed during service conditions tests.

Test	Band			
	I	J	I+J	GAR-I
Internal battery voltage range	X	X	X	X
External voltage range	X	X	X	X
Self test	X	X		X
Current	X	X	X	X
Receiver sensitivity	X	X		X
False alarm	X	X		X
Receiver bandwidth	X	X		X
Delay	X	X		X
Frequency stability	X	X		
Transmitter pulse width	X	X		
Transmitter peak power	X	X		

4.10.1 Altitude. The AN/PPN-19(V), including battery box shall be subjected to the test of MIL-STD-810C, Procedure I, Method 500.1. The AN/PPN-19(V) shall be tested at 40,000 ft (12,192m) non-operating for one hour. The operating altitude in step 2 shall be 10,000 ft (3,048m). Inability of AN/PPN-19(V) to meet the requirements of 3.11.1 shall constitute failure of this test.

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4.10.2 High temperature. The AN/PPN-19(V), including battery box shall be subjected to the test of MIL-STD-810C, Method 501.1, Procedure II. In step 7, the temperature shall be 160°F (71°C). In step 8, lower the temperature to 125°F (52°C), initiate operation of the equipment, and at the end of a 30 minute period, or when temperature stabilization occurs (whichever occurs first), the equipment shall show no degradation in performance (see 3.11.2).

4.10.3 Low temperature. The AN/PPN-19(V), including battery box, shall be subjected to the test of MIL-STD-810C, Method 502.1, Procedure I. The temperature in step 2 shall be maintained for a period of not less than 2 hours following stabilization of the test item. Following the 2 hour stabilization period, raise the chamber temperature to the low operating temperature (step 4) of -50°F (-45°C) and maintain until temperature stabilization of the test item is reached. In step 5, initiate operation of the equipment and at the end of 30 minute period or when temperature stabilization occurs (whichever occurs first) the equipment shall show no degradation in performance. (see 3.11.3).

4.10.4 Humidity. The AN/PPN-19(V), including battery box shall be subjected to the test MIL-STD-810C, Method 507.1, Procedure II. Prior to measurements, excess surface water may be removed from the equipment by wiping external surfaces only. In step 6, operation of 4.9 measurements shall be performed during the last 5 hours of the last cycle (see 3.11.4).

4.10.5 Salt fog. The AN/PPN-19(V), including battery box shall be tested in accordance with Method 509.1, Procedure I of MIL-STD-810C. Inability of the AN/PPN-19(V) to meet the requirements of 3.11.5 shall constitute failure of this test.

4.10.6 Fungus. The AN/PPN-19(V), including battery box shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810C. There shall not be abundant growth colonization on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the AN/PPN-19(V) is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the AN/PPN-19(V). Inability of the AN/PPN-19(V) to meet the requirements of 3.11.6 shall constitute failure of this test.

4.10.7 Rain. The AN/PPN-19(V) shall be tested in accordance with method 506.1, Procedure I of MIL-STD-810C. The AN/PPN-19(V) shall be operated during the last 10 minutes of each 30-minute test period. Each 30-minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period (see 3.11.7).

4.10.8 Immersion. The AN/PPN-19(V), as prepared for field transport, shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810C. This test shall be performed before and after the vibration and shock tests (see 3.11.8).

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4.10.9 Wind. Wind effect analysis shall be performed once at the beginning of the contract or at first article test only (see 3.11.9).

4.10.10 Dust. The AN/PPN-19(V) shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810C (see 3.11.10).

4.10.11 Vibration (loose cargo). The AN/PPN-19(V) excluding the battery box, shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810C. The AN/PPN-19(V), operated after the test, shall not have any degradation in performance and shall not suffer any mechanical damage (see 3.11.11).

4.10.12 Vibration (secured cargo). The AN/PPN-19(V), in its combat transportation mode and excluding the battery box, shall be subjected to the test of Method 514.2, Procedure X of MIL-STD-810C. The vibration level shall be in accordance with Fig. 514.2-7 curve AW. The cycling and sweep time shall be in accordance with table 514.2-VII. The AN/PPN-19(V), operated after the test, shall not have any degradation in performance and shall not suffer any mechanical damage (see 3.11.12).

4.10.13 Shock. The AN/PPN-19(V), in its combat transportation mode and excluding the battery box, shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810C. During part II of the test, the AN/PPN-19(V) shall be subjected to the test of Method 516.2, Procedure I, Fig. 516.2-1 of MIL-STD-810C, using a 20g's peak value. The AN/PPN-19(V), operated after the tests, shall not have any degradation in performance and shall not suffer any mechanical damage (see 3.11.13).

4.10.14 Bench handling. The AN/PPN-19(V), excluding battery box, shall be subjected to the test of Method 516.2, Procedure V of MIL-STD-810C. The AN/PPN-19(V), operated after the test, shall not have any degradation in performance and shall not suffer any mechanical damage (see 3.11.14).

4.11 Electromagnetic interference (EMI). The AN/PPN-19(V) shall be tested for compliance to the requirements of MIL-STD-461A enumerated in 3.12. The tests shall be in accordance with the approved measurement techniques of MIL-STD-462, Notice 3 (see 3.12).

4.12 Reliability. To determine compliance with 3.13, the AN/PPN-19(V) shall be tested in accordance with test Plan XV C of MIL-STD-781. Operation shall consist of applying simulated signal inputs and monitoring output signals. Each unit shall transmit a minimum of five minutes for every ten hours of operating time. The minimum test time per equipment shall be in accordance with paragraph 4.5.2.4 of MIL-STD-781. (see 3.13).

4.12.1 Reliability test conditions. Reliability testing shall be performed on equipment which have successfully completed 4.6 and 4.9.

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4.12.1.1 Electrical stress. The equipment shall be operated at nominal design input voltage for 50 percent of the time and at minimum and maximum input voltages for 25 percent of the time, respectively. Power shall be applied through the battery terminal.

4.12.1.2 Thermal stress. The equipment shall be operated at its specific ambient temperature conditions for minimum to maximum as indicated in Figure B-1 of MIL-STD-781. The following ambient temperatures shall be used:

Cold soak temperature: -57°C (-70.6°F) (start of normal cycle)

Operating temperature range: -46°C to + 68°C (-50.8°F to +154.4°F)

4.12.1.3 Vibration stress. The vibration level shall be 2.2g's ± 10 percent at a nonresonant frequency between 20 and 60 Hz. The duration of vibration shall be at least 15 minutes during each operating hour.

4.12.1.4 Moisture. Moisture levels sufficient to cause visible condensation and frosting shall be used. Humidity need not be held constant during the test cycle, and high levels may be accomplished by moisture injection at appropriate times in the test cycle.

4.12.1.5 Test cycle. The test cycle shall be twenty-four hours.

4.12.2 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification and approved test procedures. Paragraph 5.6 through 5.9 inclusive of MIL-STD-781 shall apply. The type of failures and required actions shall be in accordance with paragraph 5.6.1 of MIL-STD-781. Determination of compliance shall be in accordance with paragraph 5.7 of MIL-STD-781.

4.12.3 Parameters to be measured.

4.12.3.1 Daily performance parameters. Performance parameters are to be measured and recorded daily in accordance with paragraph 5.5.3 of MIL-STD-781. The minimum parameters to be measured daily shall be transmitter frequency and power output in both I- and J-band, BIT and undervoltage protection. A complete set of measurements shall be taken daily during both the high and low temperature stabilization. Vibration shall not be avoided during measurement.

4.12.3.2 Weekly performance parameters. Performance parameters are to be measured and recorded weekly in accordance with paragraph 5.5.2 of MIL-STD-781. The minimum parameters to be measured weekly shall be transmitter frequency and power output in both I- and J-band, BIT, and sensitivity in I- and J- and GAR-I band.

4.13 Visual and mechanical. The AN/PPN-19(V) shall be examined for the defects listed in MIL-STD-252. In addition, the system shall be visually inspected to determine conformance to the requirements of 3.6.

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4.13.1 SEL Form 1183. An inspection shall be performed, on one unit only, during First Article Testing, utilizing SEL Form 1183, to verify compliance with Para 3.6 (System Safety Engineering).

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, the first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packaging inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VII, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE VII. Packaging inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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5. PACKAGING.

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, Tables I through VII, coded as follows:

5.1.1 Level A. JT-1-00-00-NR-X-ED.

5.1.2 Level B. JS-1-00-00-NR-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES.

6.1 Intended use. The AN/PPN-19(V) is intended for use by Army ground units to provide tactical aircraft with terminal guidance for ordnance delivery or drop zone locations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.
b. Level A, or B preservation and packing (see section 5).
c. When first article inspection rough handling tests are not required.
d. Marking and shipping of samples.
e. Place of final inspection.
f. Number of first article to be submitted for approval (see 3.1).
g. When first article packaging inspection reports require acquisition activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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6.5 Subject term (key word) List.

Radar
Radar Receiver-Transmitter RT-1493/PPN-19(V)
Antenna AS-3779/PPN-19(V)
Switch, Waveguide SA-2495/PPN-19(V)
Control, Transponder Set C-11601/PPN-19(V)
Radar Transponder Beacon

Custodian:
Army-CR

Preparing activity:
Army-CR
Project 5895-A354

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-49303(CR)	Transponder Set AN/PPN-19(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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